

HDOC Day-15

Menu-Driven C Program Using switch-case

Problem Statement

Input a positive integer and use a menu-driven switch-case program to perform one of these operations:

- Choice 1: Check whether all digits of the number are even.
- Choice 2: Check whether all digits of the number are prime digits (2, 3, 5, or 7).

Algorithm

1. Start.
2. Input a positive integer n.
3. Display the menu: 1. Check all digits are even 2. Check all digits are prime.
4. Input choice.
5. Copy n into a temporary variable temp.
6. For Choice 1, examine each digit using $\text{digit} = \text{temp} \% 10$. If any digit is odd, mark the number as not all-even. Otherwise, after all digits are checked, all digits are even.
7. For Choice 2, examine each digit. A prime digit must be 2, 3, 5, or 7. If any digit is not one of these, mark the number as not all-prime.
8. Display the appropriate result.
9. For any other choice, display Invalid choice.
10. Stop.

Test Case Table

Test	Choice	Input	Expected Output	Purpose
1	1	24680	All digits are even	All digits are even
2	1	12345	All digits are not even	Contains odd digits
3	2	2357	All digits are prime	Every digit is 2, 3, 5 or 7
4	2	2379	All digits are not prime	Digit 9 is not prime

C Program

```
#include <stdio.h>

int main()
{
```

```

int n, temp, digit, choice, flag;

printf("Enter a positive integer: ");
scanf("%d", &n);

printf("\n1. Check whether all digits are even");
printf("\n2. Check whether all digits are prime");
printf("\nEnter your choice: ");
scanf("%d", &choice);

switch (choice)
{
    case 1:
        temp = n;
        flag = 1;
        while (temp > 0)
        {
            digit = temp % 10;
            if (digit % 2 != 0)
            {
                flag = 0;
                break;
            }
            temp = temp / 10;
        }
        if (flag == 1)
            printf("All digits are even\n");
        else
            printf("All digits are not even\n");
        break;

    case 2:
        temp = n;
        flag = 1;
        while (temp > 0)
        {
            digit = temp % 10;
            if (digit != 2 && digit != 3 && digit != 5 && digit != 7)
            {
                flag = 0;
                break;
            }
            temp = temp / 10;
        }
        if (flag == 1)
            printf("All digits are prime\n");
        else
            printf("All digits are not prime\n");
        break;

    default:
        printf("Invalid choice\n");
}
return 0;
}

```

Sample Execution

Choice 1 — all digits even

Enter a positive integer: 24680
Enter your choice: 1
All digits are even

Choice 1 — not all digits even

Enter a positive integer: 12345
Enter your choice: 1
All digits are not even

Choice 2 — all digits prime

Enter a positive integer: 2357
Enter your choice: 2
All digits are prime

Choice 2 — not all digits prime

Enter a positive integer: 2379
Enter your choice: 2
All digits are not prime

Screenshots

Screenshot 1: Source Code / Compilation

```
#include <stdio.h>
int main()
{
    int n, temp, digit, choice, flag;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    printf("\n1. Check whether all digits are even");
    printf("\n2. Check whether all digits are prime");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);
    switch (choice)
    {
        case 1:
            temp = n;
            flag = 1;
            while (temp > 0)
            {
                digit = temp % 10;
                if (digit % 2 != 0)
                {
                    flag = 0;
                    break;
                }
                temp = temp / 10;
            }
            if (flag == 1)
                printf("All digits are even\n");
            else
```

```
            else
                printf("All digits are not even\n");
            break;
        case 2:
```

```

case 2:
temp = n;
flag = 1;
while (temp > 0)
{
digit = temp % 10;
if (digit != 2 && digit != 3 && digit != 5 && digit != 7)
{
flag = 0;
break;
}
temp = temp / 10;
}
if (flag == 1)
printf("All digits are prime\n");
else
printf("All digits are not prime\n");
break;
default:
printf("Invalid choice\n");
}
return 0;

```

```

(kali㉿kali)-[~/Desktop]
$ vi sample11.c

(kali㉿kali)-[~/Desktop]
$ gcc sample11.c -o sample11 -lm

```

Screenshot 2: Choice 1 Output

```

(kali㉿kali)-[~/Desktop]
$ ./sample11
Enter a positive integer: 24680

1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are even

```

```

(kali㉿kali)-[~/Desktop]
$ ./sample11
Enter a positive integer: 12345

1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are not even

```

Screenshot 3: Choice 2 Output

```

(kali㉿kali)-[~/Desktop]
$ ./sample11
Enter a positive integer: 2357

1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are prime

```

```
(kali㉿kali)-[~/Desktop]  
$ ./sample11  
Enter a positive integer: 2379  
  
1. Check whether all digits are even  
2. Check whether all digits are prime  
Enter your choice: 2  
All digits are not prime
```